

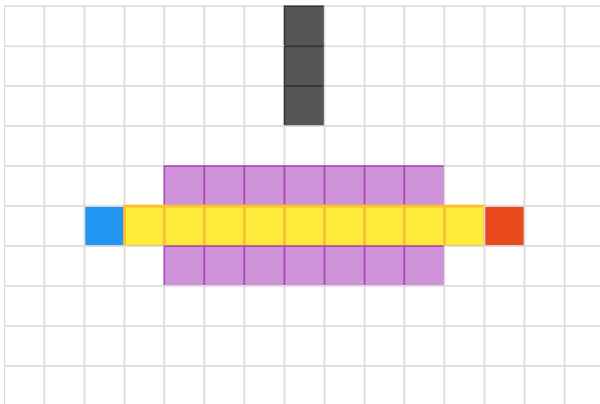


Select Lab Module:

Part 2: Terrain Optimization Lab (Swamp Grid) ▾

Reset Module

Grid Pathfinding: "The Swamp Trap"



Start Goal Wall

☐ Normal Space (**Cost: 1**) ☐ Swamp/Traffic (**Cost: 5**)

Algorithm Controls

Choose a heuristic and watch how it handles the high-cost Swamp terrain.

Select Heuristic $h(n)$:3. $h(n) = \text{Manhattan} \times 10$ (Inadmissible) ▾

Run Grid Search

Nodes Expanded (Time): 11

Total Path Cost: 38

Is Path Optimal?: **NO (Cost 38)**

Inadmissible (Manhattan $\times 10$): 🚨 The Trap! The heuristic 'pull' was so strong it ignored the heavy swamp cost. It plowed through the mud (Cost: 39). Sacrificed the shortest path!